

How to Present Your Work Without Losing the Audience

Jakub Gajarský

University of Warsaw and Masaryk University

LMW, Lisbon, 19.7.2026

How to Present Your Work Without Losing the Audience

How to give a talk

Jakub Gajarský

University of Warsaw and Masaryk University

LMW, Lisbon, 19.7.2026

How to Present Your Work Without Losing the Audience

How to (not) give a talk

Jakub Gajarský

University of Warsaw and Masaryk University

LMW, Lisbon, 19.7.2026

Who is this talk for?

Mainly for junior researchers, but could be useful for anyone.

Who is this talk for?

Mainly for junior researchers, but could be useful for anyone.

Goal:

Who is this talk for?

Mainly for junior researchers, but could be useful for anyone.

Goal:

- To make people rethink how they prepare their talks.

Who is this talk for?

Mainly for junior researchers, but could be useful for anyone.

Goal:

- To make people rethink how they prepare their talks.
- To make giving talks less stressful.

Who is this talk for?

Mainly for junior researchers, but could be useful for anyone.

Goal:

- To make people rethink how they prepare their talks.
- To make giving talks less stressful.

The use case I have in mind:

Who is this talk for?

Mainly for junior researchers, but could be useful for anyone.

Goal:

- To make people rethink how they prepare their talks.
- To make giving talks less stressful.

The use case I have in mind:

- Conference / workshop, moderate audience size (40-50 people).

Who is this talk for?

Mainly for junior researchers, but could be useful for anyone.

Goal:

- To make people rethink how they prepare their talks.
- To make giving talks less stressful.

The use case I have in mind:

- Conference / workshop, moderate audience size (40-50 people).
- Talk length of ca 15-25 minutes.

Who is this talk for?

Mainly for junior researchers, but could be useful for anyone.

Goal:

- To make people rethink how they prepare their talks.
- To make giving talks less stressful.

The use case I have in mind:

- Conference / workshop, moderate audience size (40-50 people).
- Talk length of ca 15-25 minutes.
- My bias — I work in theoretical computer science.

Part I: General considerations

Two basic principles

In any communication, the most important aspect is the **recipient of the information**. The recipient significantly affects both the content and the form of communication.

Two basic principles

In any communication, the most important aspect is the **recipient of the information**. The recipient significantly affects both the content and the form of communication.

Recipient of the information = audience

Two basic principles

In any communication, the most important aspect is the **recipient of the information**. The recipient significantly affects both the content and the form of communication.

Recipient of the information = audience

You cannot communicate anything to someone who is not paying **attention**.

What does it mean to give a successful talk?

What does it mean to give a successful talk?

Goal of giving a talk:

I want a nontrivial part of the audience to learn what we (me & my coauthors) did.

What does it mean to give a successful talk?

Goal of giving a talk:

I want a nontrivial part of the audience to learn what we (me & my coauthors) did.

What is “nontrivial part of the audience”?

What does it mean to give a successful talk?

Goal of giving a talk:

I want a nontrivial part of the audience to learn what we (me & my coauthors) did.

What is “nontrivial part of the audience”?

Of the 50 people in the audience:

What does it mean to give a successful talk?

Goal of giving a talk:

I want a nontrivial part of the audience to learn what we (me & my coauthors) did.

What is “nontrivial part of the audience”?

Of the 50 people in the audience:

- 10 are not really there.

What does it mean to give a successful talk?

Goal of giving a talk:

I want a nontrivial part of the audience to learn what we (me & my coauthors) did.

What is “nontrivial part of the audience”?

Of the 50 people in the audience:

- 10 are not really there.
- 5 are experts in your (sub-)field.

What does it mean to give a successful talk?

Goal of giving a talk:

I want a nontrivial part of the audience to learn what we (me & my coauthors) did.

What is “nontrivial part of the audience”?

Of the 50 people in the audience:

- 10 are not really there.
- 5 are experts in your (sub-)field. The talk is not primarily for them.

What does it mean to give a successful talk?

Goal of giving a talk:

I want a nontrivial part of the audience to learn what we (me & my coauthors) did.

What is “nontrivial part of the audience”?

Of the 50 people in the audience:

- 10 are not really there.
- 5 are experts in your (sub-)field. **The talk is not primarily for them.**
- The remaining 35 are the audience you should have in mind when preparing the talk.

The talk at a conference is **not** for specialists in your (sub-)field.

The talk at a conference is **not** for specialists in your (sub-)field.

Why? Because:

The talk at a conference is **not** for specialists in your (sub-)field.

Why? Because:

- If there are 50 people in the audience, it is strange to talk to only 5 of them.

The talk at a conference is **not** for specialists in your (sub-)field.

Why? Because:

- If there are 50 people in the audience, it is strange to talk to only 5 of them.
- There is a good chance that they know the result anyway.

The talk at a conference is **not** for specialists in your (sub-)field.

Why? Because:

- If there are 50 people in the audience, it is strange to talk to only 5 of them.
- There is a good chance that they know the result anyway.
- For experts, it is usually easy to see *what* your result says, and they care mostly about *how* you proved it. Non-experts typically cannot follow this.

The talk at a conference is **not** for specialists in your (sub-)field.

Why? Because:

- If there are 50 people in the audience, it is strange to talk to only 5 of them.
- There is a good chance that they know the result anyway.
- For experts, it is usually easy to see *what* your result says, and they care mostly about *how* you proved it. Non-experts typically cannot follow this.

Why care about the non-experts?

The talk at a conference is **not** for specialists in your (sub-)field.

Why? Because:

- If there are 50 people in the audience, it is strange to talk to only 5 of them.
- There is a good chance that they know the result anyway.
- For experts, it is usually easy to see *what* your result says, and they care mostly about *how* you proved it. Non-experts typically cannot follow this.

Why care about the non-experts?

- They form the majority of the audience, and they are curious about your topic.

The talk at a conference is **not** for specialists in your (sub-)field.

Why? Because:

- If there are 50 people in the audience, it is strange to talk to only 5 of them.
- There is a good chance that they know the result anyway.
- For experts, it is usually easy to see *what* your result says, and they care mostly about *how* you proved it. Non-experts typically cannot follow this.

Why care about the non-experts?

- They form the majority of the audience, and they are curious about your topic.
- The audience members are in program committees, hiring committees, grant committees etc.

The talk at a conference is **not** for specialists in your (sub-)field.

Why? Because:

- If there are 50 people in the audience, it is strange to talk to only 5 of them.
- There is a good chance that they know the result anyway.
- For experts, it is usually easy to see *what* your result says, and they care mostly about *how* you proved it. Non-experts typically cannot follow this.

Why care about the non-experts?

- They form the majority of the audience, and they are curious about your topic.
- The audience members are in program committees, hiring committees, grant committees etc. You represent (sub-)field to the general audience.

Takeaway:

One should aim to be understood by a majority people in the room and prepare the talk with this in mind.

Takeaway:

One should aim to be understood by a majority people in the room and prepare the talk with this in mind.

You might say:

“This is obvious. I and majority of people know this and prepare their talks with this in mind.”

Takeaway:

One should aim to be understood by a majority people in the room and prepare the talk with this in mind.

You might say:

“This is obvious. I and majority of people know this and prepare their talks with this in mind.”

Try the following: During the conference, look around and check

Takeaway:

One should aim to be understood by a majority people in the room and prepare the talk with this in mind.

You might say:

“This is obvious. I and majority of people know this and prepare their talks with this in mind.”

Try the following: During the conference, look around and check

- How many people follow the talk at the beginning.

Takeaway:

One should aim to be understood by a majority people in the room and prepare the talk with this in mind.

You might say:

“This is obvious. I and majority of people know this and prepare their talks with this in mind.”

Try the following: During the conference, look around and check

- How many people follow the talk at the beginning.
- How many people follow the talk after 5-7 minutes.

Takeaway:

One should aim to be understood by a majority people in the room and prepare the talk with this in mind.

You might say:

“This is obvious. I and majority of people know this and prepare their talks with this in mind.”

Try the following: During the conference, look around and check

- How many people follow the talk at the beginning.
- How many people follow the talk after 5-7 minutes.
- How many people keep the attention until the end.

Part II: Preparing a talk

Part II: Preparing a talk

- (i) Content — deciding what to include and what to leave out.
- (ii) Preparing the slides.
- (iii) Preparing the talk itself.

Part II: Preparing a talk

- (i) Content — deciding what to include and what to leave out.
- (ii) Preparing the slides.
- (iii) Preparing the talk itself.

You cannot communicate anything to someone who is not paying attention.

Part II: Preparing a talk

- (i) Content — deciding what to include and what to leave out.
- (ii) Preparing the slides.
- (iii) Preparing the talk itself.

You cannot communicate anything to someone who is not paying attention.

How this principle materializes: When making choices about your talk, avoid things that make people lose attention.

Part II: Preparing a talk

- (i) Content — deciding what to include and what to leave out.
- (ii) Preparing the slides.
- (iii) Preparing the talk itself.

You cannot communicate anything to someone who is not paying attention.

Part II: Preparing a talk

- (i) **Content** — deciding what to include and what to leave out.
- (ii) Preparing the slides.
- (iii) Preparing the talk itself.

You cannot communicate anything to someone who is not paying attention.

Content — what to include and what to leave out

Content — what to include and what to leave out

You have 20 minutes to communicate your results.

Content — what to include and what to leave out

You have 20 minutes to communicate your results.

Wrong strategy:

I have these 8 things that I want to say, how do I pack it into 20 minutes?

Content — what to include and what to leave out

You have 20 minutes to communicate your results.

Wrong strategy:

I have these 8 things that I want to say, how do I pack it into 20 minutes?

Correct strategy:

I have 20 minutes, how much can I meaningfully communicate within that time?

Content — what to include and what to leave out

I have 20 minutes, how much can I meaningfully communicate within that time?

Content — what to include and what to leave out

I have 20 minutes, how much can I meaningfully communicate within that time?

1. Think about which concepts / definitions do you assume for granted.

Content — what to include and what to leave out

I have 20 minutes, how much can I meaningfully communicate within that time?

1. Think about which concepts / definitions do you assume for granted.
2. Every nontrivial newly introduced concept that is **not properly explained** will cost you a big part of the audience.

Content — what to include and what to leave out

I have 20 minutes, how much can I meaningfully communicate within that time?

1. Think about which concepts / definitions do you assume for granted.
2. Every nontrivial newly introduced concept that is **not properly explained** will cost you a big part of the audience.

Properly explaining a new concept: Giving context, giving examples, introducing the definition step-by-step.

Content — what to include and what to leave out

I have 20 minutes, how much can I meaningfully communicate within that time?

1. Think about which concepts / definitions do you assume for granted.
2. Every nontrivial newly introduced concept that is **not properly explained** will cost you a big part of the audience.

Properly explaining a new concept: Giving context, giving examples, introducing the definition step-by-step.

This costs time.

Content — what to include and what to leave out

I have 20 minutes, how much can I meaningfully communicate within that time?

1. Think about which concepts / definitions do you assume for granted.
2. Every nontrivial newly introduced concept that is **not properly explained** will cost you a big part of the audience.

Properly explaining a new concept: Giving context, giving examples, introducing the definition step-by-step.

This costs time. You can introduce at most **2-3 nontrivial concepts per talk**.

Content — what to include and what to leave out

I have 20 minutes, how much can I meaningfully communicate within that time?

You can introduce at most 2-3 nontrivial concepts per talk.

Example

In what follows, we will speak about finite simple undirected graphs.

Example

In what follows, we will speak about finite simple undirected graphs.

Let $\mathcal{C}$ be a graph class transducible from a graph class of bounded expansion. Then there exists an algorithm with runtime $O(n^4)$ that computes...

Example

In what follows, we will speak about finite simple undirected graphs.

Let $\mathcal{C}$ be a graph class **transducible** from a graph class of **bounded expansion**. Then there exists an algorithm with runtime $O(n^4)$ that computes...

Bounded expansion

Let G and H be graphs, and assume that $V(H) = \{1, \dots, m\}$. We say that H is a **r -shallow minor of G** if there exist disjoint sets $V_1, \dots, V_m$ of radius at most r such that for every $u, v \in V(H)$ we have that if $ij \in E(G)$ then there exist vertices $u \in V_i$ and $v \in V_j$ such that $uv \in E(G)$.

Class of graphs has **bounded expansion** if there exists a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that for every r , every $G \in \mathcal{C}$ and every r -shallow minor H of G we have that

$$\frac{|E(H)|}{|V(H)|} \leq f(r)$$

Content — what to include and what to leave out

I have 20 minutes, how much can I meaningfully communicate within that time?

You can introduce at most 2-3 nontrivial concepts per talk.

Content — what to include and what to leave out

I have 20 minutes, how much can I meaningfully communicate within that time?

You can introduce at most 2-3 nontrivial concepts per talk.

What if I need to introduce more concepts to state my result?

Content — what to include and what to leave out

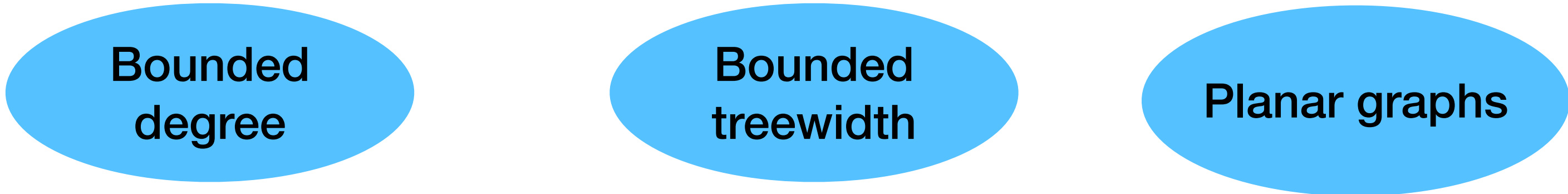
I have 20 minutes, how much can I meaningfully communicate within that time?

You can introduce at most 2-3 nontrivial concepts per talk.

What if I need to introduce more concepts to state my result?

Describe them indirectly — relate them to concepts the audience knows.

Bounded expansion



Bounded
degree

Bounded
treewidth

Planar graphs

Bounded expansion

Sparse graph
classes

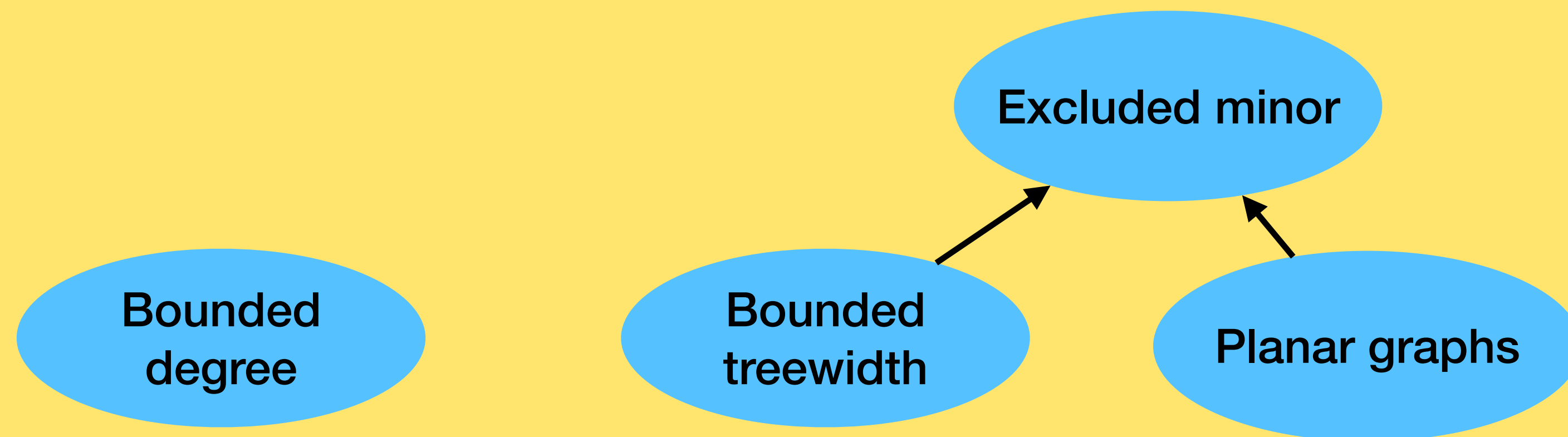
Bounded
degree

Bounded
treewidth

Planar graphs

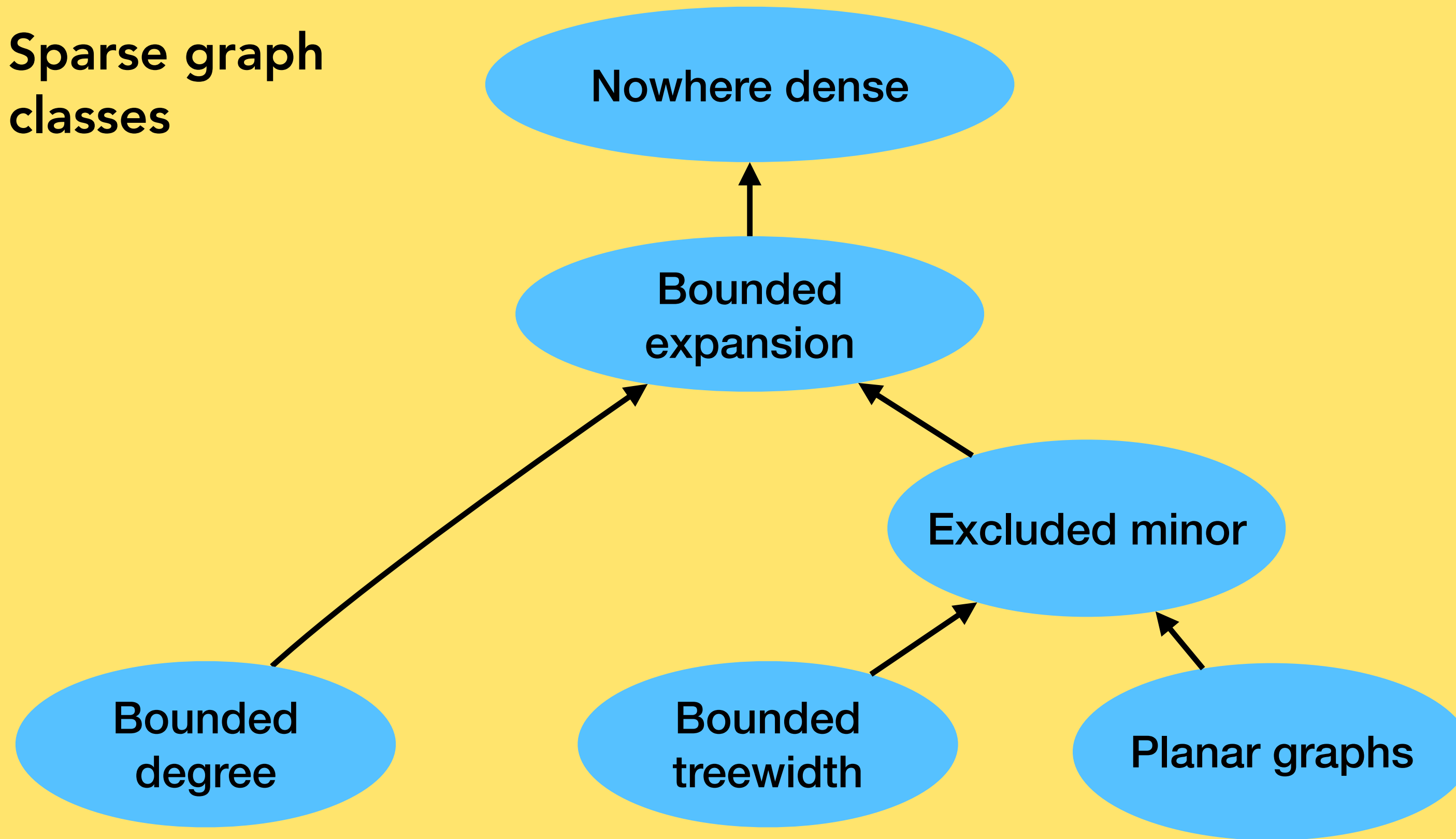
Bounded expansion

Sparse graph
classes



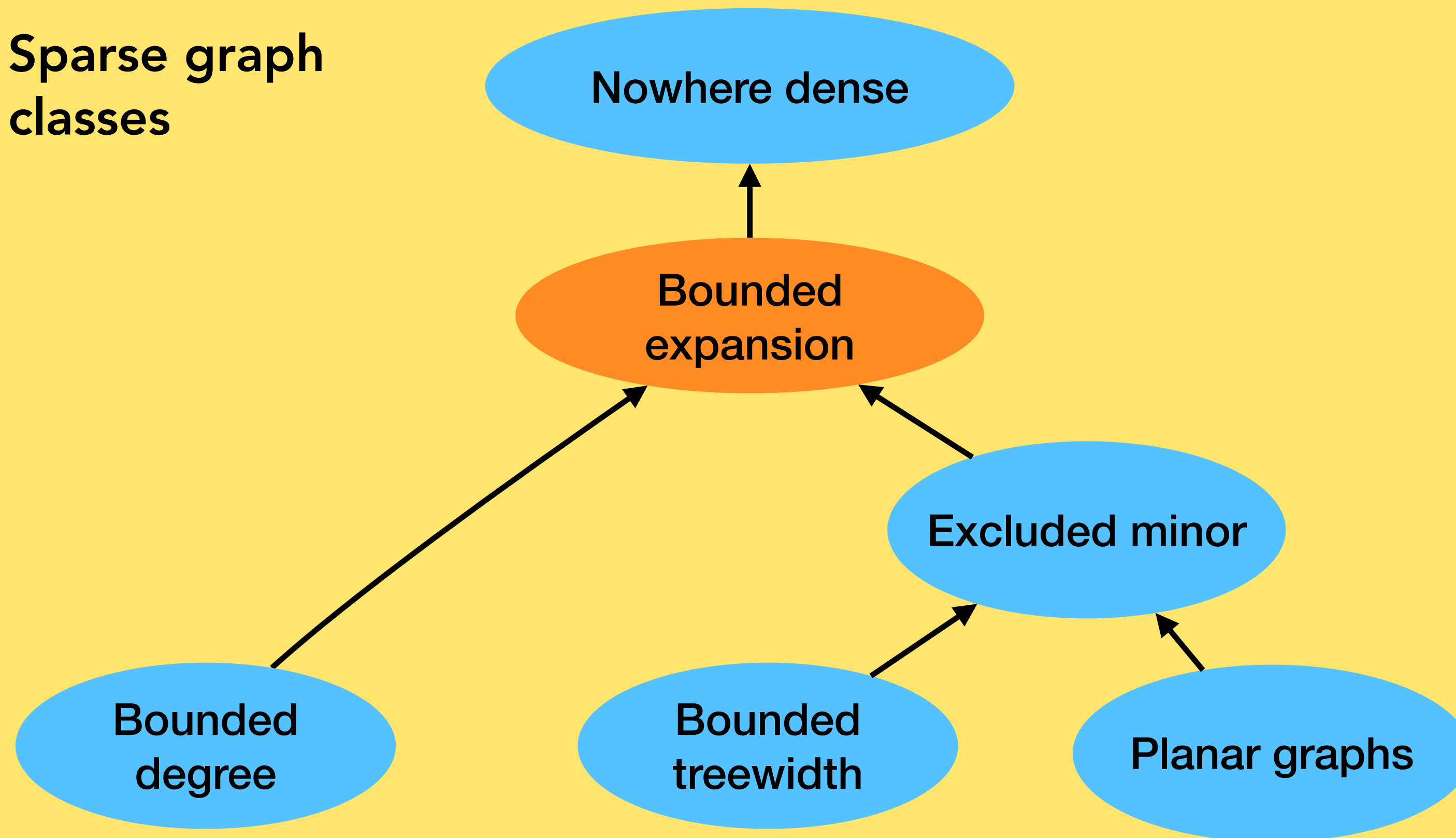
Bounded expansion

Sparse graph
classes



Bounded expansion

Sparse graph
classes



Bounded expansion:

- ▶ Very general
- ▶ Very well understood
- ▶ Many characterizations and algorithms

Example

In what follows, we will speak about finite simple undirected graphs.

Let $\mathcal{C}$ be a graph class **transducible** from a graph class of **bounded expansion**. Then there exists an algorithm with runtime $O(n^4)$ that computes...

Example

In what follows, we will speak about finite simple undirected graphs.

Let $\mathcal{C}$ be a graph class **transducible** from a graph class of **bounded expansion**. Then there exists an algorithm with runtime $O(n^4)$ that computes...

Key takeaways:

- Focus on 2-3 most important things.
- Don't be afraid to leave things out.

Part II: Preparing a talk

- (i) Content — deciding what to include and what to leave out.
- (ii) Preparing the slides.
- (iii) Preparing the talk itself.

You cannot communicate anything to someone who is not paying attention.

Part II: Preparing a talk

- (i) Content — deciding what to include and what to leave out.
- (ii) Preparing the slides.**
- (iii) Preparing the talk itself.

You cannot communicate anything to someone who is not paying attention.

Preparing slides

Standard advice:

Preparing slides

Standard advice:

1. Use pictures / illustrations whenever possible.

Preparing slides

Standard advice:

1. Use pictures / illustrations whenever possible.
2. Ideally, one idea per slide.

Preparing slides

Standard advice:

1. Use pictures / illustrations whenever possible.
2. Ideally, one idea per slide.
3. Avoid cluttered slides with a lot of text.

Preparing slides

Standard advice:

1. Use pictures / illustrations whenever possible.
2. Ideally, one idea per slide.
3. Avoid cluttered slides with a lot of text.
4. Use a large font.

Preparing slides

Standard advice:

1. Use pictures / illustrations whenever possible.
2. Ideally, one idea per slide.
3. Avoid cluttered slides with a lot of text.
4. Use a large font.

Extra advice:

Preparing slides

Standard advice:

1. Use pictures / illustrations whenever possible.
2. Ideally, one idea per slide.
3. Avoid cluttered slides with a lot of text.
4. Use a large font.

Extra advice:

- **Avoid having a static slide for too long. People get bored and lose the attention.**

Preparing slides

Standard advice:

1. Use pictures / illustrations whenever possible.
2. Ideally, one idea per slide.
3. Avoid cluttered slides with a lot of text.
4. Use a large font.

Extra advice:

- **Avoid having a static slide for too long. People get bored and lose the attention.**
- **What you say should match what appears on the slide when you say it.**

Part II: Preparing a talk

- (i) Content — deciding what to include and what to leave out.
- (ii) Preparing the slides.
- (iii) Preparing the talk itself.

You cannot communicate anything to someone who is not paying attention.

Part II: Preparing a talk

- (i) Content — deciding what to include and what to leave out.
- (ii) Preparing the slides.
- (iii) Preparing the talk itself.**

You cannot communicate anything to someone who is not paying attention.

Preparing the talk itself

Most importantly: Practice your talk before giving it.

Preparing the talk itself

Most importantly: Practice your talk before giving it.

Common trap: I have the slides prepared and I understand my stuff. I will simply explain it to the audience.

Preparing the talk itself

Most importantly: Practice your talk before giving it.

1. Break the talk into parts.

Preparing the talk itself

Most importantly: Practice your talk before giving it.

1. Break the talk into parts.
2. Try to give the talk, and mark how long each part is.

Preparing the talk itself

Most importantly: Practice your talk before giving it.

1. Break the talk into parts.
2. Try to give the talk, and mark how long each part is.
3. Notice the problematic parts and adjust them.

Preparing the talk itself

Most importantly: Practice your talk before giving it.

1. Break the talk into parts.
2. Try to give the talk, and mark how long each part is.
3. Notice the problematic parts and adjust them.
4. Repeat 2. and 3. until you are happy with the result.

Preparing the talk itself

Most importantly: Practice your talk before giving it.

1. Break the talk into parts.
2. Try to give the talk, and mark how long each part is.
3. Notice the problematic parts and adjust them.
4. Repeat 2. and 3. until you are happy with the result.

Signs of not being well prepared:

Preparing the talk itself

Most importantly: Practice your talk before giving it.

1. Break the talk into parts.
2. Try to give the talk, and mark how long each part is.
3. Notice the problematic parts and adjust them.
4. Repeat 2. and 3. until you are happy with the result.

Signs of not being well prepared:

- Handwaving or speaking about things that are not on the slide — slides not well prepared.

Preparing the talk itself

Most importantly: Practice your talk before giving it.

1. Break the talk into parts.
2. Try to give the talk, and mark how long each part is.
3. Notice the problematic parts and adjust them.
4. Repeat 2. and 3. until you are happy with the result.

Signs of not being well prepared:

- Handwaving or speaking about things that are not on the slide — slides not well prepared.
- Not finishing on time.

Preparing the talk itself

Most importantly: Practice your talk before giving it.

Crucial things about practicing the talk:

Preparing the talk itself

Most importantly: Practice your talk before giving it.

Crucial things about practicing the talk:

1. Accept that it takes time and effort.

Preparing the talk itself

Most importantly: Practice your talk before giving it.

Crucial things about practicing the talk:

1. Accept that it takes time and effort.
2. **It will take the nervousness away in two ways:**

Preparing the talk itself

Most importantly: Practice your talk before giving it.

Crucial things about practicing the talk:

1. Accept that it takes time and effort.
2. **It will take the nervousness away in two ways:**
 - **The knowledge of being prepared helps.**

Preparing the talk itself

Most importantly: Practice your talk before giving it.

Crucial things about practicing the talk:

1. Accept that it takes time and effort.
2. **It will take the nervousness away in two ways:**
 - **The knowledge of being prepared helps.**
 - **More importantly, it leads to good talks, and you'll lose the fear.**

Conclusion

Key takeaways:

Conclusion

Key takeaways:

1. Be very mindful about the audience — what do you assume that the average audience member knows?

Conclusion

Key takeaways:

1. Be very mindful about the audience — what do you assume that the average audience member knows?
2. Use attention as your guideline.

Conclusion

Key takeaways:

1. Be very mindful about the audience — what do you assume that the average audience member knows?
2. Use attention as your guideline.
3. Content: Focus on what's really important and don't be afraid to cut things out.

Conclusion

Key takeaways:

1. Be very mindful about the audience — what do you assume that the average audience member knows?
2. Use attention as your guideline.
3. Content: Focus on what's really important and don't be afraid to cut things out.
4. Slides: What you say should match what appears on a slide.

Conclusion

Key takeaways:

1. Be very mindful about the audience — what do you assume that the average audience member knows?
2. Use attention as your guideline.
3. Content: Focus on what's really important and don't be afraid to cut things out.
4. Slides: What you say should match what appears on a slide.
5. Preparation: Practice, practice, practice!